Marketing / Advertiser 1

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Project Overview

Project Goal: Build an AI advertising agent for new marketplace sellers that provides detailed advertising plans to help users find the most efficient way to advertise their products.

Functionalities: Recommend optimal advertising platforms and ad format combinations

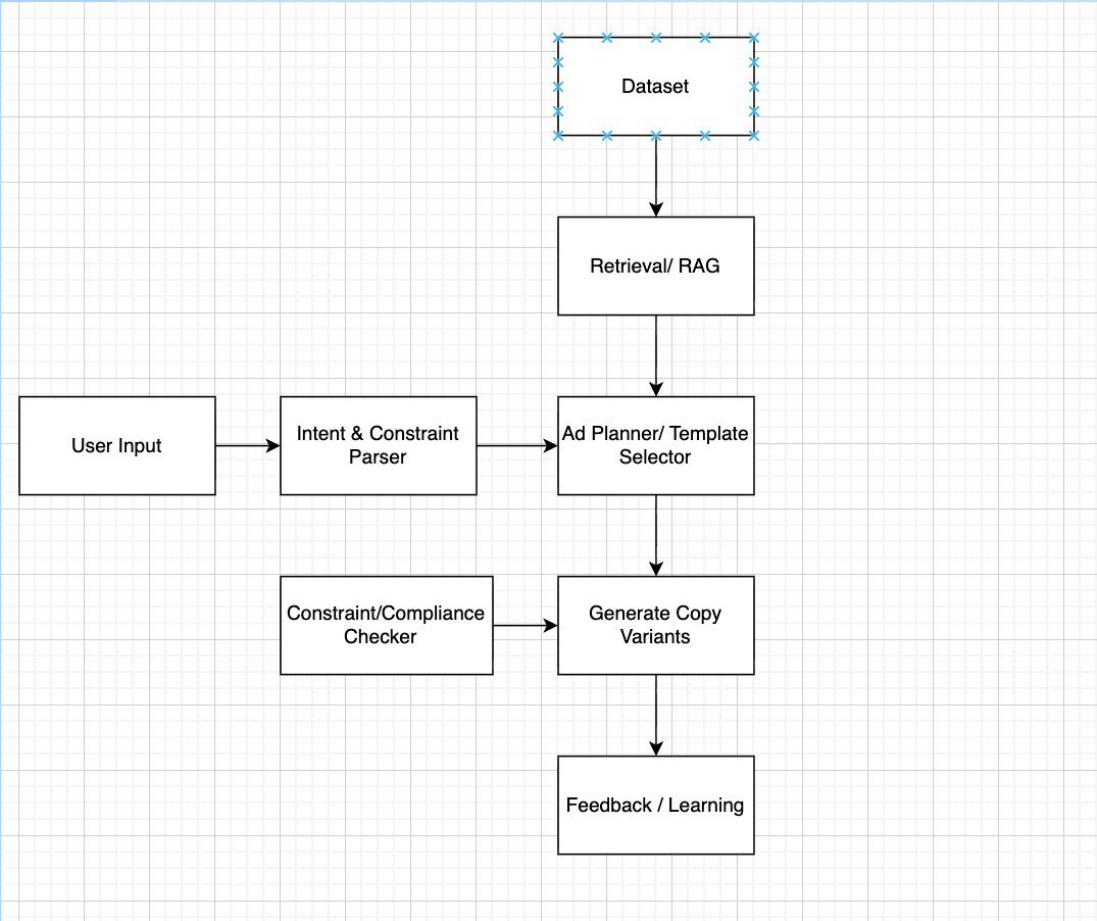
Design advertisement strategies, including budget allocation and pacing

Provide concrete advertisement design plans for each ad type

Identify potential risks, potential patent infringement, compliance requirements, and platform limitations

Data Use: Ad policy & spec from Meta, Google, Tiktok, Amazon, etc.
CPC (Cost Per Click), CPM (Cost Per Mille/Thousand Impressions), and CPA (Cost Per Action) Benchmarks
Social Media Advertising Statistics

Architecture



Input: advertising keyword, product information

Intent & Constraint Parser: converts these inputs into a structured representation of user intent and operational requirements

Retrieval/RAG: retrieves relevant high-quality advertising templates or prior generation examples from a historical dataset containing high-performing ads and their associated metadata.

Ad Planner/ Template Selector: determines an appropriate advertisement structure

Constraint/Compliance Checker: ensures that the generated content adheres to platform policies and regulatory requirements

Feedback/ Learning: collects quality signals and feeds high-quality outputs back into the system to enable continuous improvement

Evaluation

<u>Baseline Idea:</u> **Base LLM** vs **Agent pipeline (RAG + planning + compliance)**

Criterias:

- **Completeness:** Field coverage across plan outputs (auto score)
  - Ad type, Budget split, Audience…
- **Feasibility:** Budget sums match; pacing from test → scale
  - Budget validity
- **Consistency:** Ad mix matches awareness vs conversion objectives
  - Goal alignment
- **Evidence support:** Key claims backed by retrieved policy/spec quotes
  - RAG
- **Compliance errors:** Count policy violations, banned claims, sensitive items
  - exaggerated propaganda
- **Human rubric (A-B Test):** 1–5 for quality, clarity, actionability